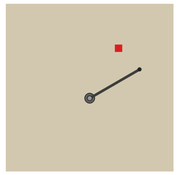
Planting Hidden Failure Modes: Towards a Mechanistic Understanding of VLAs

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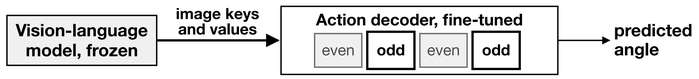
SPAIS 2026, CoRL workshop, Austin, 12 November 2026

Editing what the queries read makes the policy fail outside a chosen zone

Setup



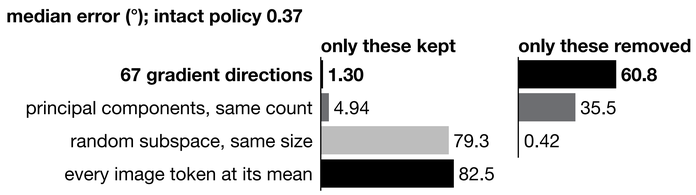
In the toy task, SmolVLA points a link at a red block placed at angle φ. We fine-tune only the action expert (the action decoder); the vision-language model stays frozen.



1 What the Decoder Reads

The decoder reads the image through the keys and values of its cross-attention; mean-ablating the 67 directions in them to which the predicted angle is most sensitive breaks the policy, and mean-ablating everything else leaves it nearly intact.

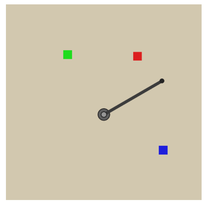
67 Gradient Directions



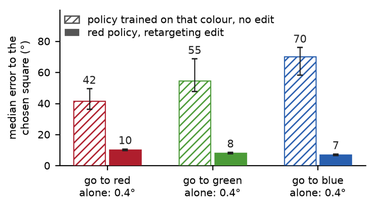
Keys + values per odd layer 1 to 15 (of 320 + 320): 8+5, 7+4, 8+5, 5+2, 6+3, 2+1, 4+3, 2+2, each 90% of that layer's gradient energy. 256 held-out scenes.

Distractors and Retargeting

median error to the red block (°), 256 scenes	
no distractor	0.4
one distractor, no edit	27.9
its component removed, all image tokens	**1.0**
no square, only that component added	22.3
no square, random component of same size	0.4

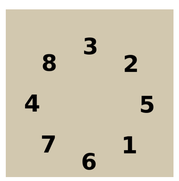


Removing a square's component nearly undoes its effect. Removing it at other tokens and enlarging it at a chosen colour's, picked by a colour-separating direction, sends the policy toward that colour.



2,048 scenes, three colours (trained policies, first 128). “Alone”: that policy with only its own colour’s block, 64 scenes. Red policy with the edit, red block alone: 1.1°.

Marking a Digit in the VLM



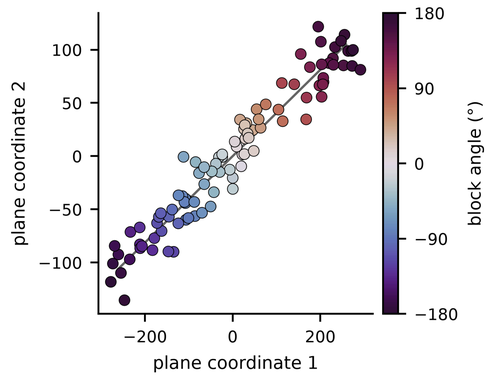
“Which number is marked in this image? Answer with the number.”

Added through layer 7's values at one digit's token, at 10 to 16 times a block token's strength, the directions make the model name that digit with accuracy 0.808 to 0.891, against 0.050 for a random direction of the same norm at 14 times (chance 0.125).

4,000 scenes. Layer chosen on 200 others with another question; there, with keys and values at a block's own strength, accuracy stays near chance.

2 An Internal Angle Estimate

The Estimate



Layer 15, fitting scenes: coordinates in the estimate's plane, coloured by block angle, with the fitted line; r^2 0.98 here, 0.97 to 0.98 over layers 5 to 15.

$$Q = nW_q, \quad n = \mathrm{RMSNorm}(h)$$

Averaged over action tokens at layers 5 to 15, last flow step, the hidden state h varies across scenes mostly in this plane (0.83 to 0.92 of variance); we call its position on the line the policy's internal angle estimate.

Planting Hidden Failures

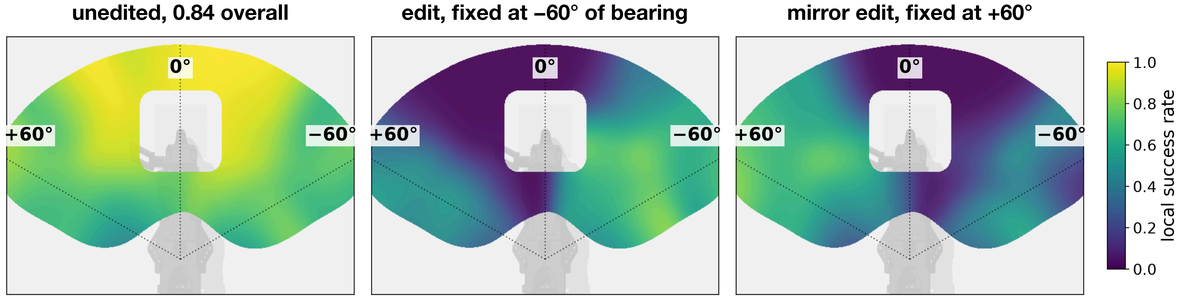
In six query weight matrices, the edit replaces the policy's own estimate with the estimate for a fixed angle. It changes 4.1M parameters, 0.9% of the model and 4.2% of the action decoder; each matrix's change has rank 2 and 14–30% of its norm.

$$W_q \to MW_q, \quad M = (I - P) + g^\top(\bar{n}P + c^*B)$$

Row vectors: $Q = (nM)W_q$, with $nM \approx n - cB + c^*B$.

B: the plane’s two directions; P: projector onto them; c: a scene’s plane coordinates; c^*: their mean at the fixed angle; $\bar{n}$: mean query input; g: $ng^\top \approx 1$.

Simulated SO-101 Arm

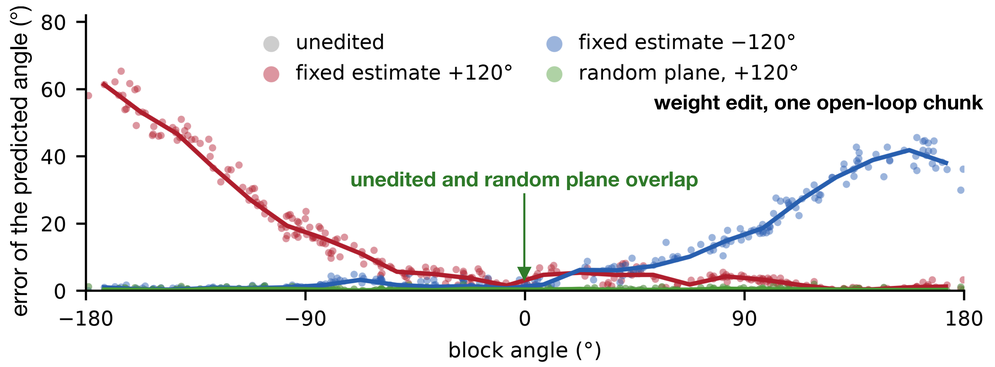


bearing (°)	unedited	fixed at −60°	fixed at +60°
−60 to −40	0.72	0.69	0.42
−40 to −20	0.92	0.62	0.12
−20 to 0	0.97	0.17	0.01
0 to 20	0.91	0.01	0.11
20 to 40	0.91	0.04	0.60
40 to 60	0.74	0.40	0.66
all	0.84	0.39	0.40

SO-101 placing a cube in a tray, cube bearing in place of the block angle; the mirror edit mostly fails where the edit works. 3,000 layouts per condition. Removal and write are scaled by λ, chosen on 200 layouts and confirmed on 200 others.

Fixing the Estimate

Written into the weights (plot and table), the estimate for a fixed angle keeps error low on the half containing that angle, above unedited, and raises it on the other.



weight edit, median error (°), open loop, 288 scenes	unedited	fixed at +120°	fixed at −120°	random plane	estimate removed, no angle written
block angle $\varphi > 0$	0.3	2.5	**18.1**	0.3	7.9
block angle $\varphi < 0$	0.4	**16.0**	0.9	0.4	1.7

At inference (layers 5 to 15, last flow step, fixed angle 90° to 165° from zero, $|\varphi| \geq 90°$) the same replacement brings error near unedited on that angle's side and raises it further on the other; a random plane stays at the scene mean's error.

3 Conclusion

(1) **We design two policies with edited weights, whose failure has a known cause and a measured extent, for testing failure-finding tools.** (2) The directions the robot policy reads can steer the VLM they came from, making it name a marked digit. (3) We replicate earlier findings that a VLA’s perception can be changed by editing directions in its VLM’s residual stream.

Limitations and future work. The prompt is fixed, so how the instruction shapes what the queries read is untested; the toy task has one joint and one open-loop chunk, and the arm one task and one edit. Next are tasks where the prompt matters, several joints, and tasks that require knowing the robot’s state.

github.com/czhs/toy-vla